

Synaptic and Dendritic Integration

The neuron's input-output function

Computational Neuroscience

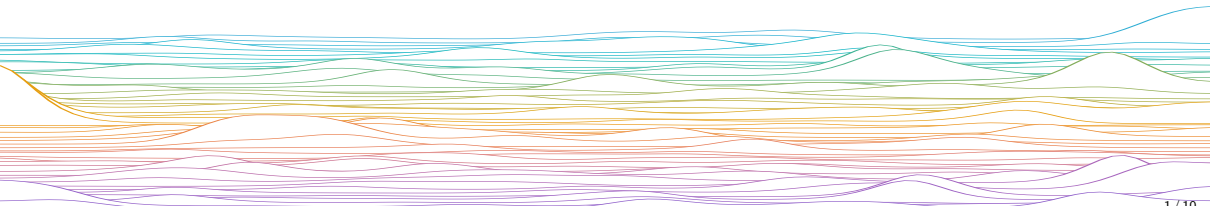
University of Bristol

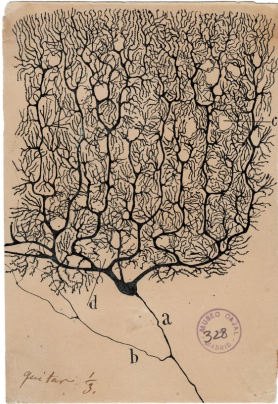
Slides adapted from C O'Donnell

M Rule

Learning outcomes:

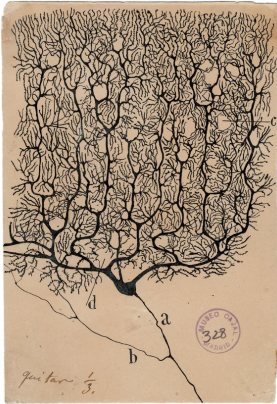
- ▶ Describe input summation in the synapse and dendrite
- ▶ Explain how input location on the dendrites affects summation
- ▶ What are some functions of dendritic nonlinearities?





Cajal

What goes on here?



Cajal

What goes on here?

Nonlinear synaptic integration

- ▶ Many spike trains in $\rightarrow$ one spike train out
- ▶ “Point” neuron models (e.g. LIF) assume soma performs a weighted linear sum of synaptic currents.

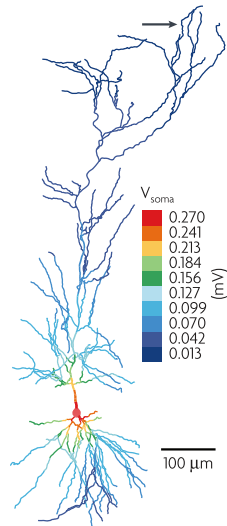
Real neurons are different!

- 1 Dendrites $\Rightarrow$ *spatial layout of synaptic inputs*.
- 2 Dendrites have voltage-dependent (active) ion channels which makes *synaptic integration nonlinear*.

Synaptic location matters

On the right is a trace of the dendritic tree of a CA1 pyramidal cell from a rat.

- ▶ Colour: amplitude of the voltage response (EPSP) at soma when the synapse is placed at the corresponding location on the dendritic tree.
- ▶ Without boosting, a synapse would give a smaller somatic response if it was located at a distal dendritic site.
- ▶ Voltage-dependent ion channels in dendrites can boost synaptic inputs to amplify their effect at the soma.

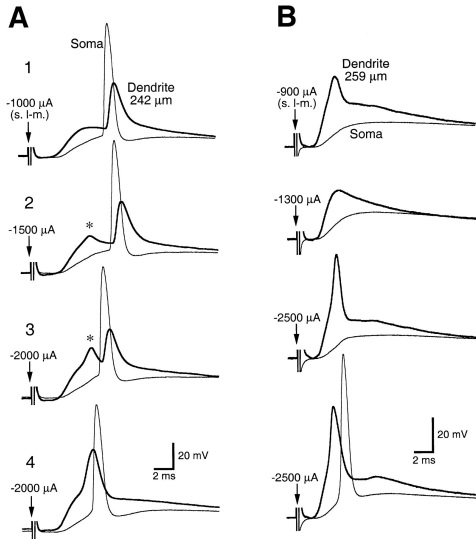


Spruston, Nat Rev Neurosci (2008)

Dendritic spikes

Some neurons have enough voltage-gated ion channels in their dendrites to enable dendritic action potentials.

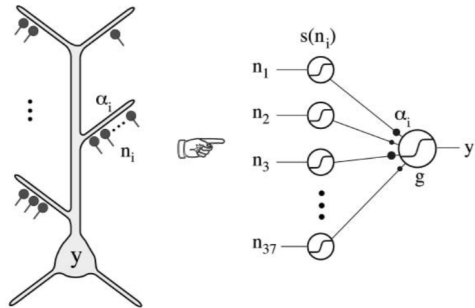
- ▶ Dendritic spikes tend to be weaker, less all-or-none than axonal action potentials
- ▶ (note variable dendritic amplitudes in each plot on right).
- ▶ One dendritic spike is not always sufficient to trigger an axonal spike.



The single neuron as two-layer neural network

Dendritic spikes $\Rightarrow$

- ▶ One pyramidal neuron $\approx$ multi-layer neural network.
- ▶ Each dendritic does a nonlinear operation on its inputs before passing the signal to the soma.
- ▶ Voltage-gated ion channels expand the brain's computational power.



Poirazi and Mel, Neuron (2003)

Summary

- ▶ Ion channels are protein complexes that let electrical current flow in and out of the neuron.
- ▶ They mediate neurons' electrical dynamics.
- ▶ Kinetic models of neurons are commonly described using ODEs.
- ▶ Ion channels' voltage dependence makes the neuron's input-output function highly nonlinear.
- ▶ Single neurons can in principle approximate a 2-layer artificial neural network

end

... Quantum Chemistry, Molecular dynamics

Molecules

Gillespie, Master Equation

Concentrations

Mass-Action Kinetics

Conductance Models

Hodgkin-Huxley

Spiking Models

Leaky Integrate and Fire

Rate Neurons

Neural Mass/Field Models

Poisson Neurons

Generalized Linear Models

Binary Neurons

McCulloch-Pitts, Hopfield, Perceptron

Cognitive Neuroscience, Psychology ...

Physiological,
Quantitative

Biological
Realism,
Data needed
to identify
parameters



Computational
Efficiency,
Mathematical
Tractability



Phenomenological,
Qualitative